

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No- 9	Team No- 6	Project Title- Dairy & Milk Cooperative Management System
------------------	-------------------	--

1. Abstract

The dairy and milk cooperative sector, particularly at the village and mandal level, continues to rely heavily on manual registers for recording milk collection, fat/SNF testing, farmer payments, and inventory of dairy products. This manual process is time-consuming, error-prone, and lacks transparency, often leading to disputes over quantity, quality-based pricing, and delayed payments to farmers. The proposed Dairy and Milk Cooperative Management System is a web-based application designed to digitize and streamline the end-to-end operations of a milk cooperative society. The system allows staff to record daily milk collection per farmer along with fat and SNF percentages, automatically calculate quality-based rates, and generate accurate payment statements. It maintains farmer profiles, tracks cooperative membership, manages inventory of milk and dairy by-products, and generates reports on collection trends, revenue, and outstanding dues. An admin dashboard provides real-time analytics on daily and monthly collection volumes and cooperative performance, while role-based access differentiates administrators, collection center staff, and farmers. The front end is built using React for a responsive and interactive interface, the back end using Node.js with Express for RESTful APIs and business logic, and a relational SQL database for structured storage of farmer, collection, and transaction records. By automating rate calculation, record-keeping, and reporting, the system aims to improve transparency, reduce manual errors, and enable data-driven decision-making for dairy cooperative societies, ultimately benefiting both farmers and management.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520030561	SURISSETTI PREM SAI	
2520030179	NITISH CHOWDHARY	
2520030284	JASWANTH REDDY	

3. Frameworks / Platforms / Software Used

(List the frameworks, libraries, platforms, or tools you plan to use.)

Frontend: React.js, HTML5, CSS3, Bootstrap/Tailwind CSS, Axios. Backend: Node.js, Express.js, JWT-based authentication, REST APIs. Database: MySQL / PostgreSQL. Tools & Platforms: Git & GitHub, Postman, Visual Studio Code, npm, Vercel/Render (deployment).

Guide Name and Signature (remarks if any) _____